Siva Meenakshi Annapureddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.27
SRI CHAITANYA JUNIOR COLLAGE , KANURU, VIJAYAWADA	Class XII (Board of Intermediate Education,Andhra pradesh)	2023	96.60%
SRI CHAITANYA GIRLS EM H S KAKKALAPALLI	Class X (Board of Secondary Education,Andhra pradesh)	2021	100.00%

POSITIONS OF RESPONSIBILITY

Sunshine Buddy : Provided peer support and a safe space for students to discuss personal and campus-related concerns.

Sunshine Mentor : Mentored first-year students in academic, social, and emotional adjustment while organizing engagement activities.

Public Relations & Networking Manager, E-Cell: Led outreach for E-Merge and E-Summit, connecting students, entrepreneurs, and industry professionals.

Public Relations & Marketing Coordinator, TEDx : Managed event promotions and audience engagement to strengthen outreach and event visibility.

WORK EXPERIENCE

WiSH'25 Mentee, Texas Instruments

May 2025 - June 2025

- Gained hands-on exposure to analog and digital circuit design, Verilog RTL, RF, and ADC/DACs.

- Performed circuit simulations and worked on design thinking project.

Analog Validation Intern, Texas Instruments

Jan 2026 - July 2026

- Power Variation Debug for Wi-Fi Microcontroller Devices: Investigated sources of RF power variation through device and board-level analysis.

- Design and Simulation of an Octagonal Resonator for PCB Material Characterization: Designed and simulated an octagonal resonator for characterizing PCB material properties at 5-7 GHz.

PROJECTS

FLARE - Smart & Sustainable Street Lighting | Design Thinking Project

- Conducted user research and problem analysis to identify safety and energy-efficiency requirements for adaptive street lighting; evaluated alternatives and defined a smart, autonomous lighting solution.
- Developed and prototyped a radar and LDR based adaptive lighting system with solar-grid hybrid power, achieving a projected 61.5% reduction in power consumption through demand-based dimming.

Air Quality Prediction using Machine Learning

- Developed a multi-city air quality prediction framework using MLP, RNN, LSTM, and XGBoost, with temporal, lag, and rolling-statistical features from pollutant and meteorological data.
- Performed XGBoost hyperparameter tuning and comparative evaluation across six cities, achieving a reported 35.8% reduction in MAE over baseline XGBoost.

Adaptive Noise Cancellation using Recursive Least Squares (RLS)

- Designed an adaptive noise cancellation system using an 8th-order RLS filter to estimate and suppress noise from noisy audio signals.
- Implemented partial noise suppression using a second-order notch filter and evaluated performance through SNR and spectral/tonal preservation metrics, achieving 35.8 dB SNR for RLS.

Power Variation Debug for Wi-Fi Microcontroller Devices, Intern - Texas Instruments

- Investigated sources of RF output power variation through device-level characterization and board-level analysis.

- Analyzed power, impedance, and frequency-dependent behavior to isolate contributors to observed variation.

Design and Simulation of an Octagonal Resonator for PCB Material Characterization, Intern - Texas Instruments

- Designed and simulated an octagonal resonator for extracting PCB material properties in the 5-7 GHz frequency range.

- Analyzed resonant frequency and quality factor to characterize dielectric and loss variations across PCB samples.

SKILLS

Programming

Hardware & Design Tools:

RELEVANT COURSES

Programming & Data:

Introduction to Programming, Data Structures & Applications, Artificial Intelligence, Machine Learning

Liberal Arts & Social Sciences:

Psychology of Interpersonal Relationships, Fundamentals of Organizational Structure, Science, Technology, and Society, Work, Technology and Society: Assembly Lines to Algorithms, Communication Skills